

Pin Diagram of 8086

- The 8086 is a 16-bit microprocessor available in 40-pin DIP (Dual Inline Package) chip. This microprocessor operates in single processor or multiprocessor configurations to achieve high performance.
- The pin configuration of 8086 is shown in the figure. Some of the pins serve a particular function in minimum mode (single processor mode) and others function in maximum mode (multiprocessor mode).

The 8086 signals are categorized into 3 types:

1. Common signals for both minimum mode and maximum mode.
2. Special signals which are meant only for minimum mode
3. Special signals which are meant only for maximum mode

40 pins can be categorized as:

- a. Supply Pins (3)
- b. Clock related pins (3)
- c. Address and Data pins (21)
- d. Interrupt pins (2)
- e. Other Control pins (3)
- f. Mode multiplexed pins- Min and Max mode (8)

a)Supply Pins (3) :

It uses 5V DC supply at VCC pin 40, and uses ground (VSS) pin 1 and 20 for its operation

b)Clock related pins (3)

CLK : Clock input:

Clock signal is provided through Pin-19. It provides timing to the processor for operations. Its frequency is different for different versions, i.e. 5MHz, 8MHz and 10MHz. The clock input signal provides the basic timing input signal for processor and bus control operation. It is asymmetric square wave with 33% duty cycle. (External clock generator 8284 is used to generate clock signal)

Ready

It is available at pin 22. It is an acknowledgement signal from I/O devices that data is transferred. It is an active high signal. When it is high, it indicates that the device is ready to transfer data. When it is low, it indicates wait state.

RESET: It is available at pin 21 and is used to restart the execution. It causes the processor to immediately terminate its present activity. When 8086 reset, it restarts the execution from

memory location FFFF0H (as CS = FFFF H and IP = 0000 H). The reset signal is active high and must be active for at least four clock cycles.

c) Address and Data pins (21)

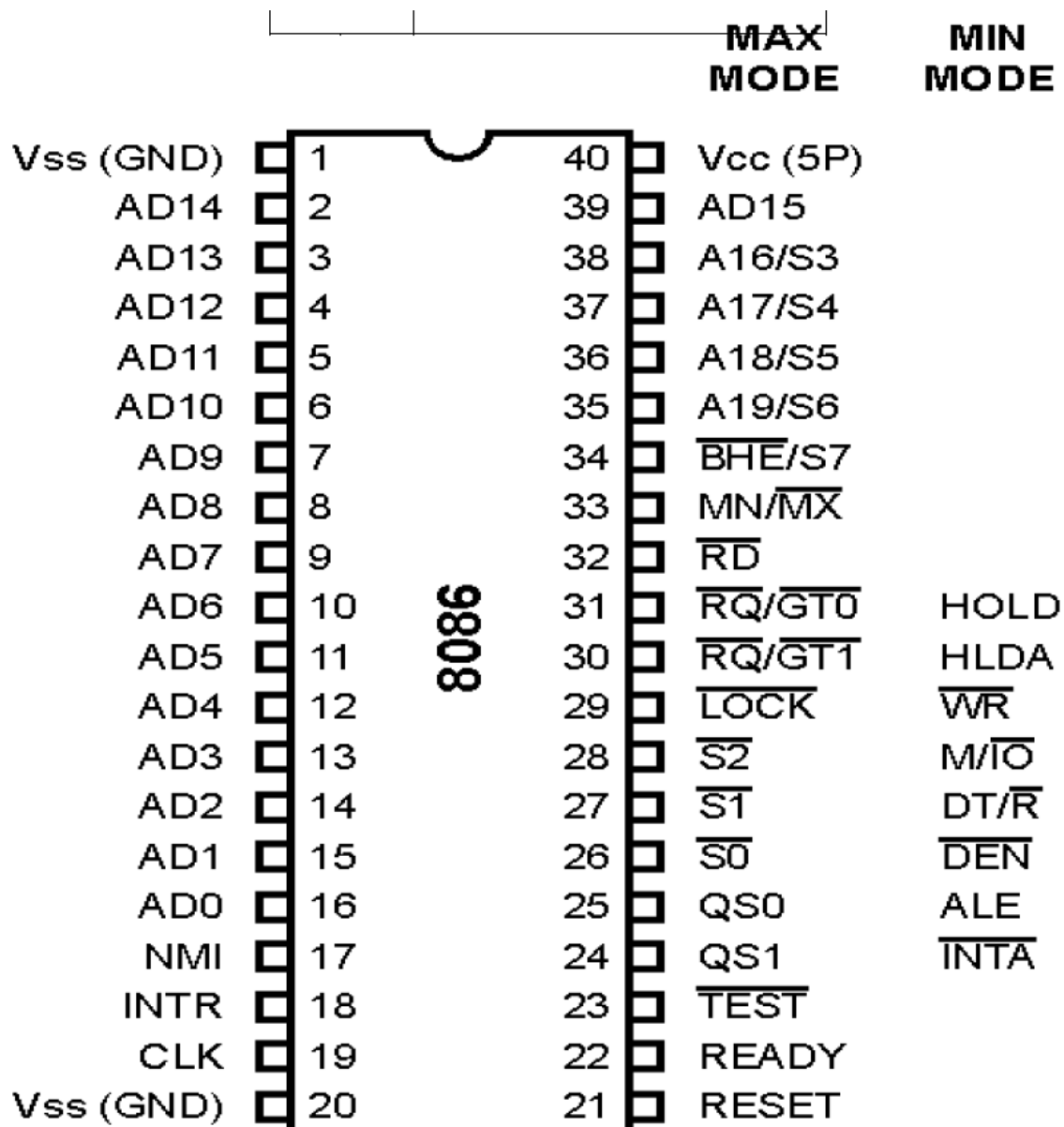
AD7 – AD0 : The address/ data bus lines are the multiplexed address data bus and contain the right most eight bit of memory address or data. The address and data bits are separated by using *ALE* signal.

AD₁₅ – AD₈ : The address/data bus lines compose the upper multiplexed address/data bus. This lines contain address bit $A_{15} - A_8$ or data bus $D_{15} - D_8$. The address and data bits are separated by using *ALE* signal.

A₁₉ / S₆ – A₁₆ / S₃ The address/status bus bits are multiplexed to provide address signals $A_{19} - A_{16}$ and also status bits $S_6 - S_3$. The address bits are separated from the status bits using the ALE signals. The status bit S_6 is always a logic 0, bit S_5 indicates the condition of the interrupt flag bit. The S_4 and S_3 indicate which segment register is presently being used for memory access.

S4	S3	Segment
0	0	Extra Segment
0	1	Stack Segment
1	0	Code Segment (or No Segment)
1	1	Data Segment

$\overline{BHE}$ / S7 : The bus high enable (BHE) signal is used to indicate the transfer of data over the higher order ($D_{15} - D_8$) data bus. It goes low for the data transfer over $D_{15} - D_8$ and is used to derive chip select of odd address memory bank or peripherals.



d) Interrupt pins (2)

INTR (Interrupt Request:)

It is available at pin 18. It is an interrupt request signal, which is sampled during the last clock cycle of each instruction to determine if the processor considered this as an interrupt or not. Interrupt request is used to request a hardware interrupt of *INTR* is held high when interrupt enable flag is set, the 8086 enters an interrupt acknowledgement cycle after the current instruction has completed its execution. It is a maskable, Level triggered interrupt.

NMI :It stands for non-maskable interrupt and is available at pin 17. It is an edge triggered input, which causes an interrupt request to the microprocessor. If NMI is activated, the interrupt input uses interrupt vector 2.

e) Other Control pins (3)

$\overline{\text{TEST}}$ (pin 23): It is active low input line dedicated for 8087 co-processor. In minimum mode it is connected to GND. In Max mode whenever co-processor is busy it makes this pin HIGH. This input is examined by "WAIT" instruction. If the TEST input goes low (when co-processor has completed its operation); execution will continue. Else the processor remains in an idle state.

MN /MX (pin 33) : The minimum/maximum mode signal to select the mode of operation either in minimum or maximum mode configuration. Logic 1 indicates minimum mode.

$\overline{\text{RD}}$ (Pin 32): Low on this line indicates read operation from memory or I/O device.

e) Mode multiplexed pins- Min and Max mode (8)

HOLD (Pin 31-minimum mode): The hold input request a direct memory access (DMA). If the hold signal is at logic 1, the micro process stops its normal execution and places its address, data and control bus at the high impedance state.

HLDA (pin 30-minimum mode): Hold acknowledgement indicates that 8086 has entered into the hold state.

$\overline{\text{RQ}}/\overline{\text{GT}}1$ (pin 30) and **$\overline{\text{RQ}}/\overline{\text{GT}}0$** (pin 31) (*Maximum mode*)- request/Grant: The request/grant pins are used by other local bus masters to force the processor to release the local bus at the end of the processors current bus cycle. These lines are bidirectional and are used to both request and grant a DMA operation. $\overline{\text{RQ}}/\overline{\text{GT}}0$ is having higher priority than $\overline{\text{RQ}}/\overline{\text{GT}}1$

$\overline{\text{LOCK}}$ (pin 29-maximum mode) : The lock output is used to lock peripherals off the system, i.e, the other system bus masters will be prevented from gaining the system bus.

$\overline{\text{WR}}$ (pin 29- minimum mode): whenever the write signal is at logic 0, the data bus transmits the data to the memory or I/O devices connected to the system.

$\overline{\text{DEN}}$ (pin 26-minimum)Data Enable: Data bus enable signal indicates the availability of valid data over the address/data lines.(enable data buffers IC 8286)

DT / $\overline{\text{R}}$ (pin 27 –minimum mode): Data transmit/Receive: This output signal is used to decide the direction of data flow through the bidirectional buffer.

Logic 1 on this line Indicates transmitting and logic 0 indicates receiving the data.

$\overline{M} / \overline{IO}$ (pin 28-minimum mode)- Memory/IO signal selects either memory operation or I/O operation. This line indicates that the microprocessor address bus contains either a memory address or an I/O port address. Signal high at this pin indicates a memory operation

$\overline{S_2}, \overline{S_1}, \overline{S_0}$ (pins 26,27,28 – maximum mode) - Status lines: These are the status lines that reflect the type of operation being carried out by the processor. These status lines are encoded as follows.

$\overline{S_2}$	$\overline{S_1}$	$\overline{S_0}$	Function
0	0	0	Interrupt Acknowledge
0	0	1	Read I/O port
0	1	0	Write I/O port
0	1	1	Halt
1	0	0	Code Access
1	0	1	Read memory
1	1	0	Write memory
1	1	1	Passive (In active)

ALE (pin 25-minimum mode)- Address Latch Enable: This output signal indicates the availability of valid address on the address/data bus, and is connected to latch enable input of latches.

$\overline{INTA}$ (pin 24-minimum mode)- Interrupt acknowledge: The interrupt acknowledge signal is a response to the INTR input signal. The INTA signal is normally used to gate the interrupt vector number onto the data bus in response to an interrupt request.

QS₀ and QS₁ (pin 25, Pin 24 – maximum mode)- Queue status: The queue status bits shows the status of the internal instruction queue. The encoding of these signals is as follows:

QS1	QS0	Function
0	0	No operation, queue is idle
0	1	First byte of opcode
1	0	Queue is empty
1	1	Subsequent byte of opcode